

Pivot: Imagery-vs-Rest Decode and RSA

Current analysis (imagery vs rest), captured so collaborators can track the findings without re-running. Full write-up: docs/pivot-analysis.md.

Why the encoder representation. EEGPT's reusable value is its encoder embedding, the vector downstream tasks freeze and build on, and the only part braindecode ships. Trust in the model comes from whether that representation encodes meaningful signal, so interpreting it is the point. The reconstruction decoder is training machinery discarded at deployment.

Findings (n=20). Decoding: EEGPT decodes imagery vs rest about as well as CSP within subject and best of three across subjects. RSA: after Autoreject cleaning and a non-motor control band, EEGPT's embedding geometry correlates with band-power broadly but not specifically with mu (mu is about equal to the control band; beta is modestly higher).

Interpretation. EEGPT decodes motor imagery competitively and transfers across subjects, but its representation is spectrally non-specific, organized by aggregate signal structure more than the mu/beta rhythms clinicians rely on. This is a trust-relevant interpretability finding. A likely explanation not yet confirmed is that the correspondence reflects overall signal power.

Decode (chance 50%)

method	within %	cross %
EEGPT embeddings	73.1	58.3
CSP+LDA	76.3	54.7
band-power+LDA	63.3	57.3

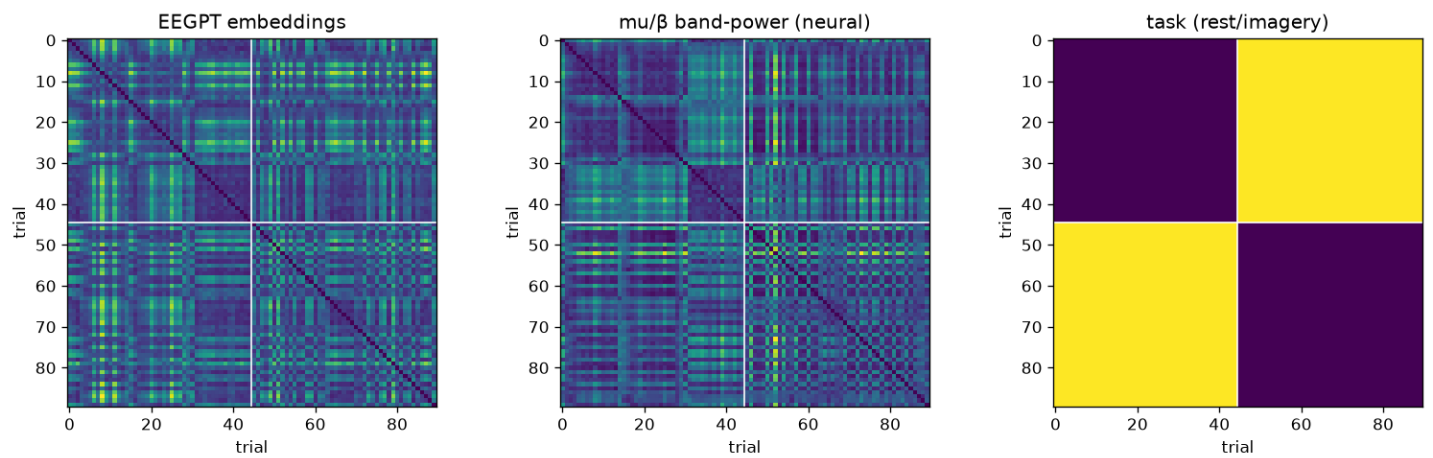
Band-resolved RSA

band	model-neural	partial (task removed)
mu (8-13)	0.204	0.183
beta (13-30)	0.243	0.228
control (2-7, non-motor)	0.202	0.169

Figures

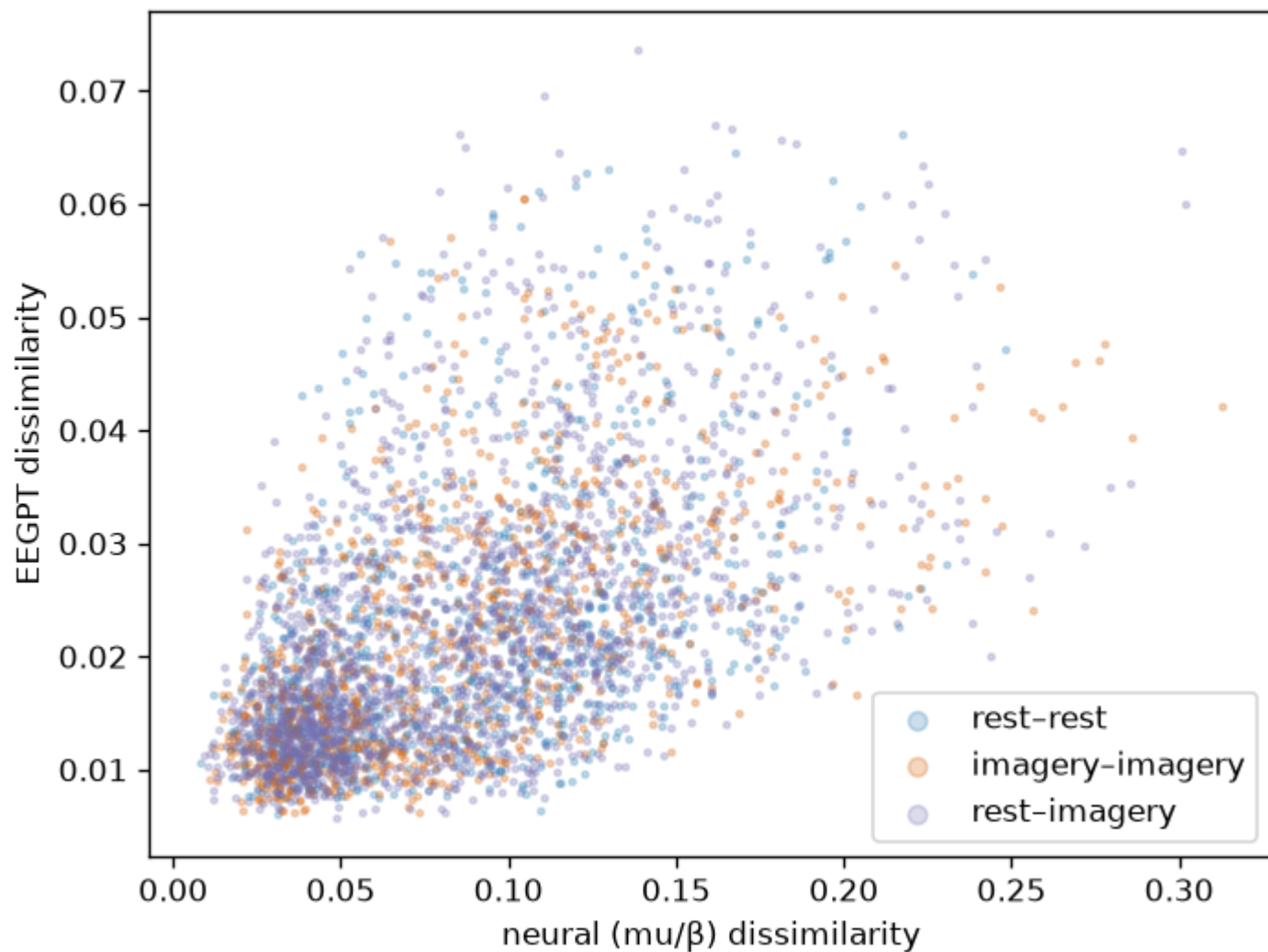
Dissimilarity maps: EEGPT vs mu/beta vs task (illustrative subject)

Dissimilarity maps (subject 6) — trials sorted: rest block, then imagery block



Each dot is a trial pair, colored by pair type (rest-rest, imagery-imagery, mixed)

Each dot = a trial pair (subject 6)
overall $r=0.58$



Decode accuracy: EEGPT vs classical, within and cross subject

Decode imagery-vs-rest

